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Kasa 92.1 FM climbs McCarthy Hill: new mast, solar backup and a signal that reaches Kasoa

Ablekuma Kasa Community Radio (AKCR) · Dansoman, Accra · Ref. AKCR/PR/2026-07 · Background facts and engineering notes for editors attached

Ablekuma Kasa Community Radio will move its 92.1 MHz transmitter from the rooftop in Dansoman to a 45-metre mast on McCarthy Hill, Weija, on Monday 19 October 2026 at 06:00, and will keep broadcasting through power cuts on a new solar-charged battery bank.

The announcement

For seventeen years Kasa 92.1 FM has broadcast from a 30-metre steel mast bolted to the roof of the old Dansoman community library. Beyond the Densu River the signal fades, and every major power cut has taken the station off air within minutes. "Our listeners in Kasoa and Bortianor kept phoning to say they lost us halfway through the morning market prices," said station manager **Naa Adjeley Quaye-Mensah**. "From 19 October they should not have to."

The new site is a fenced plot leased from the McCarthy Hill Residents' Welfare Committee for 15 years at GHS 9,600 a year. It holds a new 1,000-watt transmitter, a four-bay dipole antenna, a 12 kWp solar array on the equipment-shelter roof and a 48-volt lithium battery bank. The equipment was supplied and installed by Tetteh-Bruce Broadcast Services Ltd of Tema under contract TBB/AK/0417.

Test nights before the switch

Engineers will run test transmissions from McCarthy Hill on the nights of 5 to 15 October 2026, Monday to Thursday only, between 00:30 and 04:30. During those hours listeners may hear tone, a looped station identification or short silences. Normal programmes continue from Dansoman outside the test hours. The Dansoman rooftop transmitter stays in place as a standby until **30 November 2026** and will then be dismantled.

What listeners need to do

Nothing, for most people: the frequency stays at 92.1 MHz and no retuning is needed. Car radios with a stored preset and phones with an FM receiver will pick up the new signal automatically. Listeners in tall buildings in Dansoman may find a west-facing window gives better indoor reception.

Engineering notes for editors

Chief engineer **Kofi Ankrah-Lamptey** has supplied the three planning formulas the station uses. The figures quoted in this release were calculated with them.

$$P_{ERP} = 10 log_{10} P_{TX} + G_{ant} - L_f - L_c \qquad (1)$$

where P_{ERP} is the effective radiated power in dBW, P_{TX} the transmitter output in watts, G_{ant} the antenna gain in dBd, L_f the feeder cable loss in dB and L_c the loss in connectors and the filter in dB. To convert back to watts, ERP = $10^{P/10}$.

At McCarthy Hill the transmitter delivers 1,000 W into a four-bay antenna with a gain of 6.2 dBd. The 65 m of 7/8-inch feeder loses 2.1 dB per 100 m, which rounds to 1.4 dB, and the filter and connectors lose 0.3 dB. Equation (1) gives 30.0 + 6.2 − 1.4 − 0.3 = **34.5 dBW, about 2,820 W ERP**. The station's community broadcasting authorisation caps ERP at 3.0 kW (34.8 dBW), so the new site works just inside the limit. The Dansoman rooftop delivered 300 W into a 3.0 dBd antenna with 0.9 dB of losses, or 26.9 dBW (about 490 W ERP).

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$$d = 4.12\left(\sqrt{h_t} + \sqrt{h_r}\right) \quad (2)$$

where d is the radio horizon in kilometres, h_t the height of the antenna centre above average surrounding terrain in metres, h_r the receiving antenna height in metres (1.5 m for a hand-held or car radio, 10 m for a rooftop aerial), and 4.12 a constant that allows for normal bending of VHF waves in the atmosphere.

How far the signal reaches

The antenna centre on McCarthy Hill stands 112 m above average terrain, against 38 m at Dansoman. For a car radio (h_r = 1.5 m), equation (2) gives 4.12 × (10.58 + 1.22) = **48.6 km** from the hill, compared with 30.4 km from the old rooftop. The horizon is an upper limit: buildings and hills shorten it, so the station plans its service on a smaller primary service area with a radius of 35 km. It takes in Kasoa, Bortianor, Weija, Amasaman and the Kaneshie market; beyond it, in places such as Nsawam, reception is not guaranteed.

Staying on air when the power goes

Until now a diesel generator took up to four minutes to start after a power cut. At McCarthy Hill the transmitter, studio-to-transmitter link and exciter run from the battery bank whenever grid power fails, and the solar array recharges it during the day. The generator moves to the hill as a last reserve and starts only when the battery falls to 30 % of capacity.

$$t = \frac{C \times V \times D \times \eta}{P_{load}} \quad (3)$$

where t is the battery autonomy in hours, C the bank capacity in ampere-hours (400 Ah), V the nominal bank voltage (48 V), D the allowed depth of discharge (0.70, so the generator reserve of 30 % is never used), η the inverter efficiency (0.92) and P_{load} the total load in watts.

Worked example. At full power the site draws 2,050 W (1,800 W transmitter, 250 W link, exciter and fan). Equation (3) gives 400 × 48 × 0.70 × 0.92 ÷ 2,050 = 12,364.8 ÷ 2,050 = **6.0 hours** on battery alone. If a power cut lasts longer than three hours, the duty engineer switches the transmitter to reduced-power mode (500 W output), in which the whole site draws 1,150 W.

Reporting reception after the move

The engineering team collects reception reports from 19 October to 30 November 2026. To send one:

1. Tune to 92.1 MHz and score what you hear from 1 (nothing) to 5 (as clear as the studio).
2. Note your GhanaPost GPS digital address or the nearest landmark, and whether you were indoors, outdoors or in a vehicle.
3. Send an SMS or WhatsApp message to 024 921 0921 in the form *SIGNAL score address time*, for example SIGNAL 2 GA-544-1290 07:15.
4. If you score 1 or 2 inside the 35 km primary service area, a volunteer signal walker will call you back within three working days to arrange a visit.

Reports sent during the October test nights are logged but answered only after 19 October.

Paying for the move

The project cost GHS 512,400 in total. The Adaklu Civic Media Fund gave a grant of GHS 300,000, the Ablekuma Traders' Association contributed GHS 40,000 and GHS 100,050 came from the station's own reserves. The remainder was raised by the **Hold the Signal listener drive**, which ran from 1 June to 31 August 2026 and collected GHS 72,350 from 3,118 donors.

Background facts. AKCR was founded in 2009 and is run by 64 volunteers and 5 paid staff. It broadcasts 19 hours a day (05:00 to 24:00) in Ga, Twi, Ewe and English, with a Hausa hour on Fridays at 20:00.

Media contact: Selasi Agbenyega, press coordinator · press@kasa921.org.gh · 030 296 4410 · Dansoman Community Library, 2nd floor. Site interviews: 8 and 9 October 2026, 10:00 to 13:00.

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